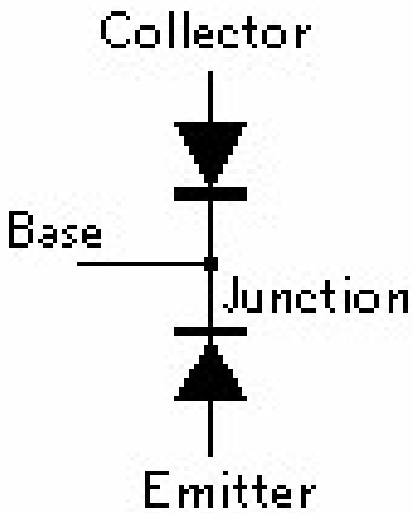
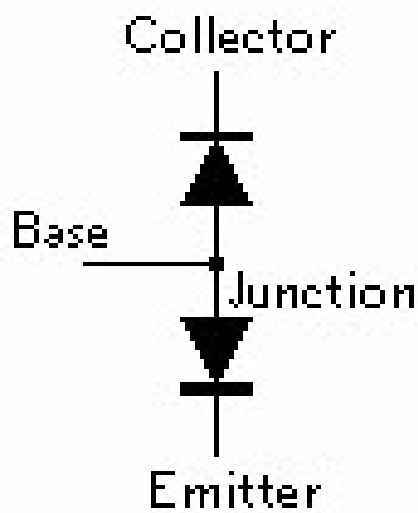
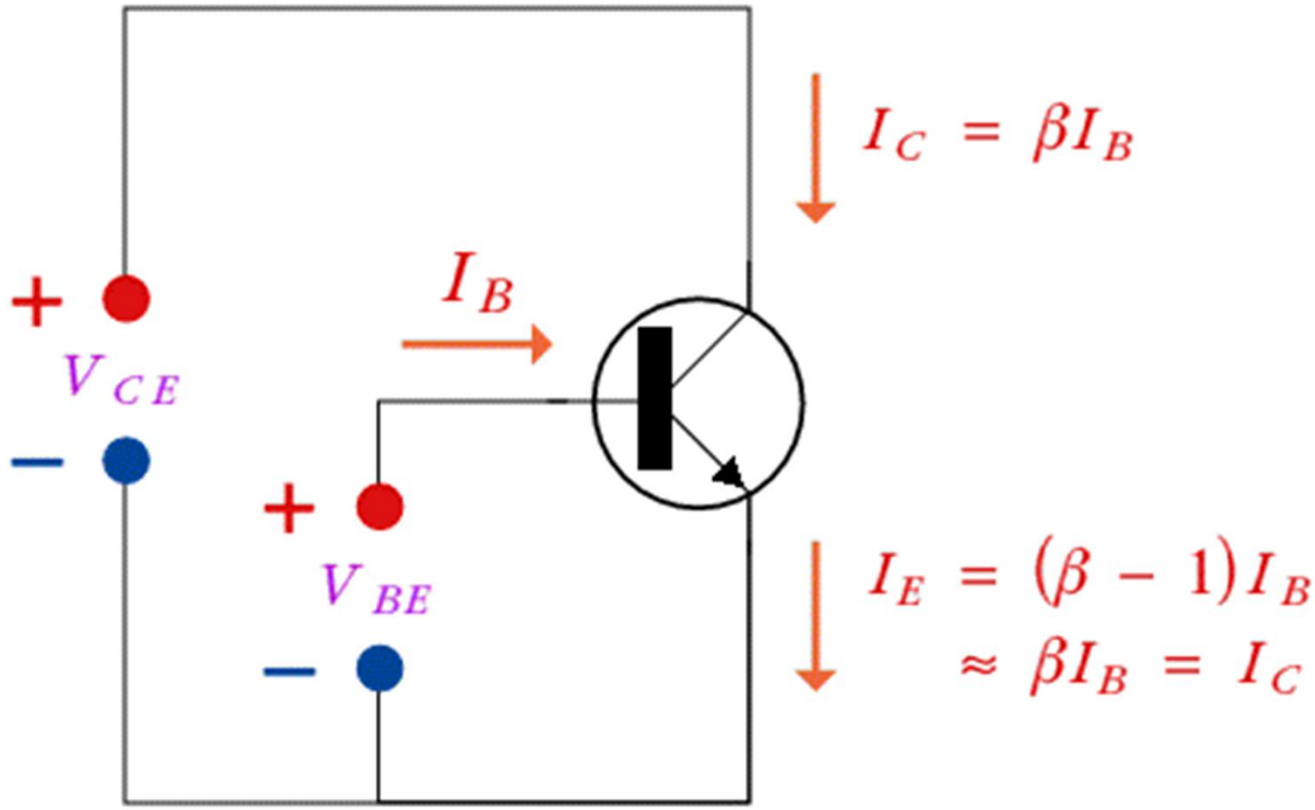
Bipolar Transistor



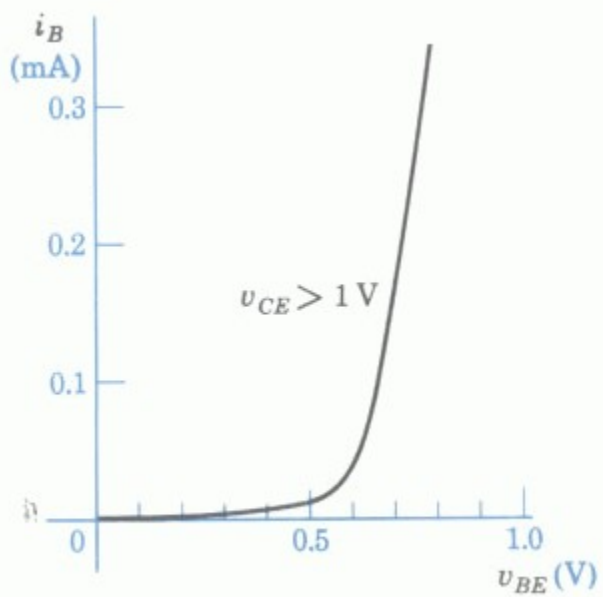
PNP Transistor
Equivalent Circuit



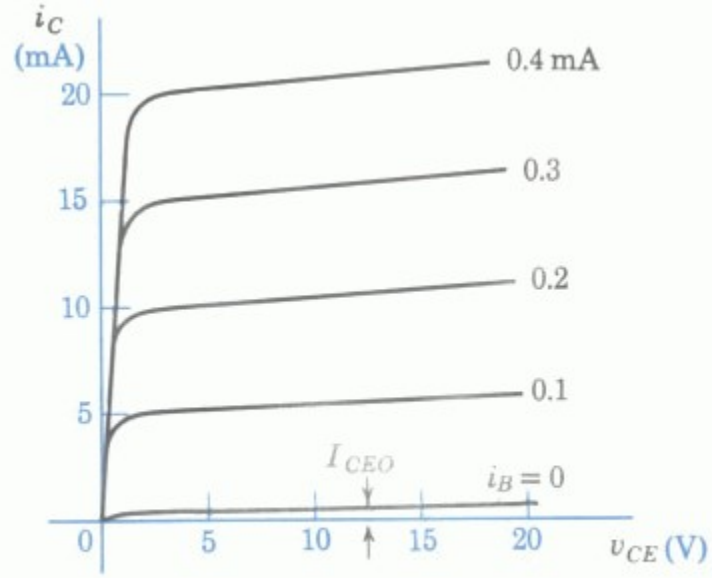
NPN Transistor
Equivalent Circuit



'Conventional' view of the NPN Bipolar Transistor



(a) Base characteristics



(b) Collector characteristics

TRANSISTOR AS A SWITCH

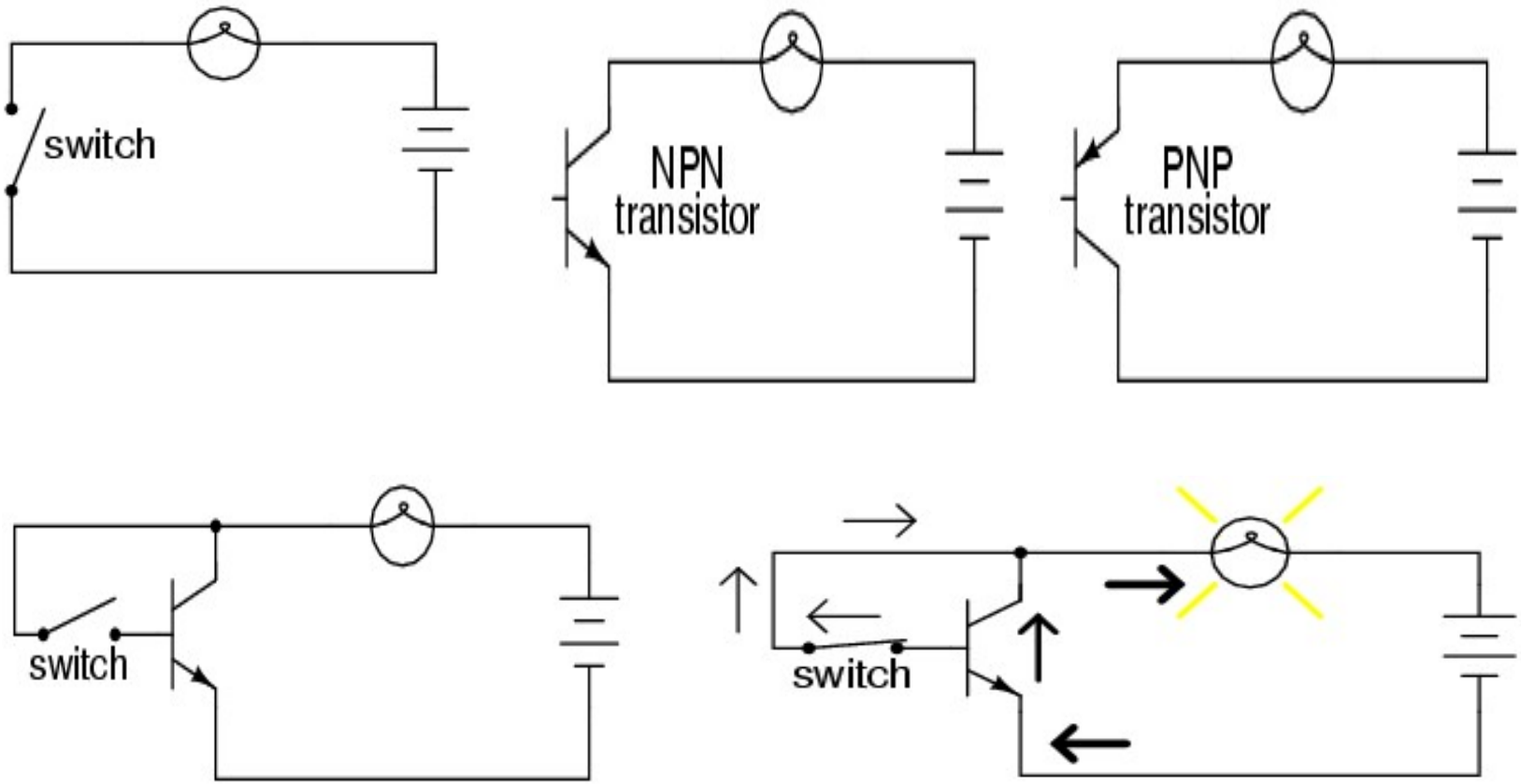
Transistors may be used as switching elements to control DC power to a load.

The switched (controlled) current goes between emitter and collector, while the controlling current goes between emitter and base.

When a transistor has zero current through it, it is said to be in a state of *cutoff* (fully non conducting).

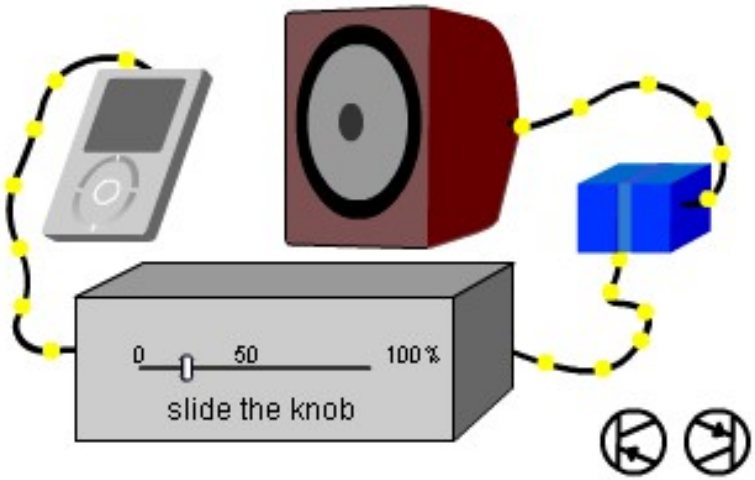
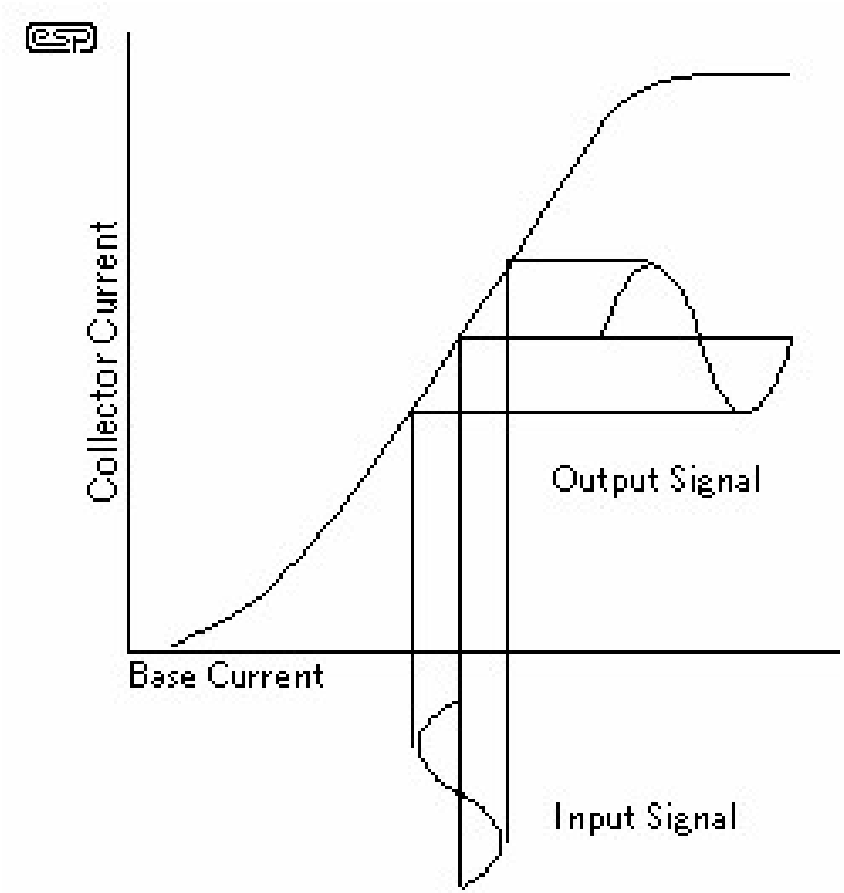
When a transistor has maximum current through it, it is said to be in a state of *saturation* (fully conducting).

TRANSISTOR AS A SWITCH



TRANSISTOR AS AN AMPLIFIER

A transistor can also amplify current. A low-level voltage (represented by the headphone output of the mp3 player), connected to the transistor's base through the volume control, will regulate the larger current powering the loudspeaker. But be careful--turn it up too high, and the transistor will be overloaded and may overheat.



TRANSISTOR AS AN AMPLIFIER